

Cache Performance Analysis: Associativity and Prefetching Effects

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1 Overview

This project investigates how cache associativity and prefetching influence CPU cache performance under different memory access patterns. A custom trace-driven cache simulator written in C models an L1 cache with configurable set-associative mapping, write-back/write-allocate policy, and LRU replacement. A fully associative shadow cache classifies misses into compulsory, conflict, and capacity categories, enabling separation of workload-driven behaviour from structural cache limitations. A configurable next-line prefetcher measures usefulness and pollution across distances 0–16.

Experiments across sequential, strided, and random workloads show that moving from direct-mapped to 4-way associativity reduces conflict misses by over 80%, from 2,863 to 543 in the sequential workload, while further increases to 256-way give diminishing returns. Prefetching at distance 1–2 consistently improves hit rate across all workloads and configurations, but pollution rises to above 60% of total prefetches beyond distance 4, eventually reducing hit rate below the no-prefetch baseline at distance 16.

2 Introduction

Context and motivation Modern CPU performance is heavily constrained by the latency gap between processor and main memory. Cache hierarchies lower this gap by exploiting temporal and spatial locality.

However, cache efficiency is not solely determined by capacity. It is also strongly influenced by placement policy (associativity), replacement strategy, and memory mechanisms such as prefetching.

Related work Classic computer architecture literature [1] shows that increasing cache associativity reduces conflict misses by providing greater flexibility in block placement, while LRU replacement exploits temporal locality effectively and remains one of the most widely used practical replacement policies [4]. Experimental evaluation of associativity [2] demonstrates that relatively modest associativity levels, such as 4-way, capture most of the available reduction in conflict misses, with diminishing returns moving forward. Hardware prefetching can improve performance for workloads exhibiting strong spatial locality, although overly aggressive prefetching may increase cache pollution by displacing useful cache contents [3].

Objectives

- Quantify miss behaviour using a shadow fully-associative baseline model
- Evaluate associativity impact on conflict miss reduction
- Analyse prefetch trade-offs using usefulness vs pollution metrics

3 Data and Experimental Setup

Simulator description A custom cache simulator written in C was developed to model L1 cache behaviour. The simulator implements a set-associative cache architecture with write-back and write-allocate policies.

A key design choice is the use of a Least Recently Used (LRU) replacement policy, implemented via timestamp tracking per set. Each cache line stores its last access time, and eviction selects the least recently accessed line within a set. It models temporal locality in a simple, deterministic way while staying computationally efficient.

Simulation complexity and performance considerations The cache simulator operates with associative scans per-access, resulting in $O(A)$ lookup complexity per cache set, where A is the associativity level. LRU replacement is implemented using timestamp-based metadata, introducing constant-time updates but requiring linear eviction search within each set. The fully associative shadow cache introduces an additional $O(N)$ lookup cost per miss classification, where N is the total number of blocks in the shadow structure. Despite these costs, the simulator maintains deterministic behaviour suitable for trace-based evaluation, with memory overhead scaling linearly with cache capacity and associativity. This cost is acceptable because trace-driven simulation prioritises correctness over throughput. Although lookup cost increases with associativity, the largest configuration remained practical for trace-driven evaluation because cache sizes are relatively small.

Miss classification model To enable deeper analysis of cache inefficiencies, a fully associative shadow cache is used as a reference model. This model represents an idealised cache with maximal associativity and identical capacity constraints.

Miss types are classified as follows:

- **Compulsory miss:** first-time access to a memory block (tracked via seen-block table)
- **Conflict miss:** block would exist in fully associative model but is evicted in set-associative cache
- **Capacity miss:** miss occurs even in fully associative model

This classification enables separation of structural cache limitations from workload behaviour.

Prefetching model A distance-based prefetcher issues speculative memory fetches for the next N cache lines following a miss. Each prefetched line is tracked using variables to measure:

- Prefetch usefulness (whether it is later accessed)
- Prefetch pollution (whether it evicts useful data before being used)

Validation methodology To ensure correctness of the simulator, cache behaviour was validated against expected theoretical trends. Increasing associativity monotonically reduced conflict misses across all configurations, and compulsory misses remained invariant across configurations, confirming correct implementation of miss classification logic. These sanity checks provide confidence that observed performance trends reflect accurate cache modelling rather than inconsistencies in the simulation. The random workload showed higher baseline hit rates than sequential and strided, which is discussed in Section 4 as a trace composition effect rather than a validity problem.

Workload	Associativity	Hit Rate (%)	Conflict Misses	Capacity Misses
Sequential	1-way	79.98	2863	1432
	4-way	85.01	543	1269
	256-way	85.39	409	1215
Strided	1-way	78.47	2643	1359
	4-way	83.59	539	1185
	256-way	84.06	373	1140
Random	1-way	87.92	2753	1452
	4-way	90.78	588	1287
	256-way	91.03	443	1233

Table 1: Hit rate and miss classification by workload and associativity (prefetch distance = 0).

4 Exploration and Analysis

Effect of cache associativity Increasing associativity reduces conflict misses by allowing more flexible block placement within cache sets. Table 1 shows hit rates and conflict miss counts across all three workloads and associativity levels with no prefetching.

Moving from direct-mapped to 4-way associativity reduces conflict misses by approximately 80% across all workloads from 2,863 to 543 in the sequential case, while capacity misses remain relatively stable, confirming that conflict misses are the dominant structural inefficiency at low associativity. Further increasing associativity to 256-way shows only small additional improvement, suggesting that LRU combined with 4-way associativity already captures most exploitable temporal locality.

Effect of prefetching Table 2 shows hit rates for the sequential workload across all prefetch distances and associativity levels.

Associativity	p=0	p=1	p=2	p=4	p=8	p=16
1-way	79.98	82.85	83.47	83.06	82.48	79.98
4-way	85.01	88.00	88.75	88.86	87.66	82.83
256-way	85.39	88.51	89.39	90.06	89.00	82.15

Table 2: Sequential workload hit rate (%) by associativity and prefetch distance.

For direct-mapped, hit rate peaks at distance 2 (83.47%) before declining, falling back to the no-prefetch baseline by distance 16 (79.98%). For 4-way and fully associative caches, the peak shifts to distance 4 (88.86% and 90.06% respectively), demonstrating that higher associativity extends the effective prefetch range by providing additional cache ways that reduce the likelihood of evicting useful blocks.

Workload sensitivity An unexpected finding is that the random workload achieves consistently higher baseline hit rates than sequential and strided workloads across all associativity levels. This is attributable to the composition of the memory trace: Valgrind’s Lackey tool captures all memory accesses during program execution, including library initialisation and runtime code with inherently sequential structure. These accesses constitute a significant proportion of the trace and exhibit strong locality, raising the effective hit rate independent of the test program’s access pattern. This highlights a limitation of trace-driven simulation using whole-program traces rather than isolated workload regions.

Prefetch efficiency vs pollution trade-off Figure 1 shows the prefetch usefulness and pollution trade-off for the 4-way sequential configuration. At distance 1, usefulness is 58% while pollution is only 13%,

giving a clear net benefit. The two metrics cross at distance 4, beyond which pollution dominates. By distance 8, usefulness falls to 21% while pollution reaches 62%, and by distance 16, usefulness is just 11% while pollution reaches 78%. This pattern is consistent across all workloads and associativity levels, and is consistent with prior work on stream buffers [3].

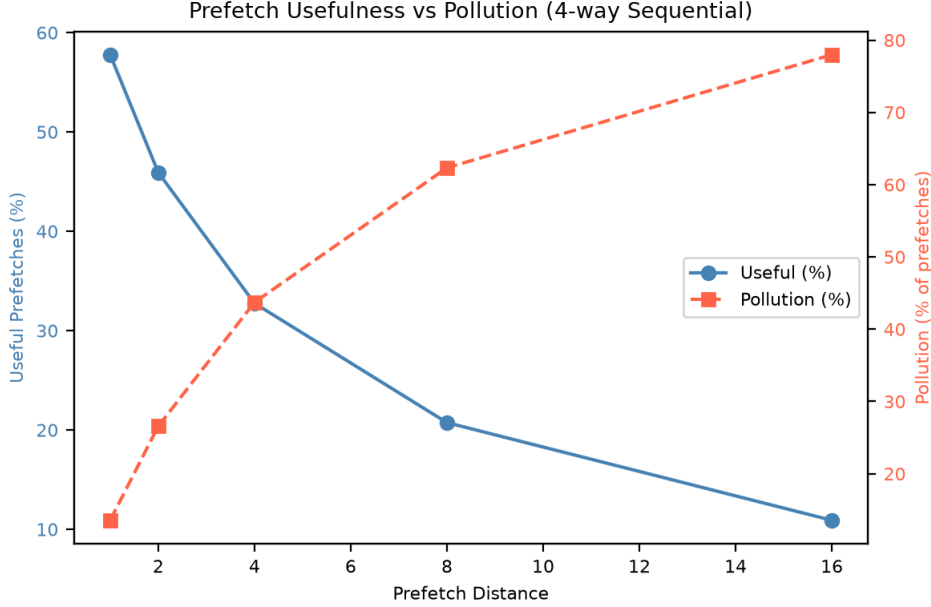


Figure 1: Prefetch usefulness vs pollution for the 4-way sequential workload. The two metrics cross at approximately prefetch distance 3–4, beyond which pollution exceeds usefulness.

Correlation analysis Figure 2 shows Pearson correlations between cache performance metrics across all 54 configurations. Hit rate shows a strong negative correlation with both conflict misses ($r = -0.81$) and total misses ($r = -0.81$), confirming that conflict misses are the primary driver of performance loss. Prefetch pollution correlates strongly with prefetch total ($r = 1.00$) and prefetch useful ($r = 0.91$), reflecting that higher prefetch aggressiveness increases both useful and wasted fetches simultaneously. Capacity misses show weaker correlations with most metrics ($r = -0.35$ with hit rate), consistent with the finding that capacity is driven by workload footprint rather than cache configuration.

5 Discussion and Conclusions

Summary of findings Cache performance is strongly determined by associativity and prefetch distance, but both show clear diminishing returns. Moving from direct-mapped to 4-way associativity reduces conflict misses by approximately 80% and improves hit rate by around 5 percent across all workloads. Further increases to 256-way yield under 1 percentage point of additional improvement. Prefetching at distance 1–2 consistently improves hit rate, with usefulness between 47% and 60% and pollution below 20%. Beyond distance 4, pollution dominates, exceeding 60% at distance 8 and 77% at distance 16, and hit rate falls below the no-prefetch baseline in direct-mapped configurations.

Crossover point A key finding is that associativity shifts the prefetch crossover point. The distance at which prefetching begins to hurt more than it helps. For direct-mapped caches, hit rate returns to baseline by distance 16. For 4-way and fully associative caches, hit rate remains above baseline even at distance 8, as additional cache ways accommodate prefetched blocks without displacing useful data. This interaction between associativity and prefetch aggressiveness suggests that prefetch distance should be tuned relative to cache associativity rather than treated independently.

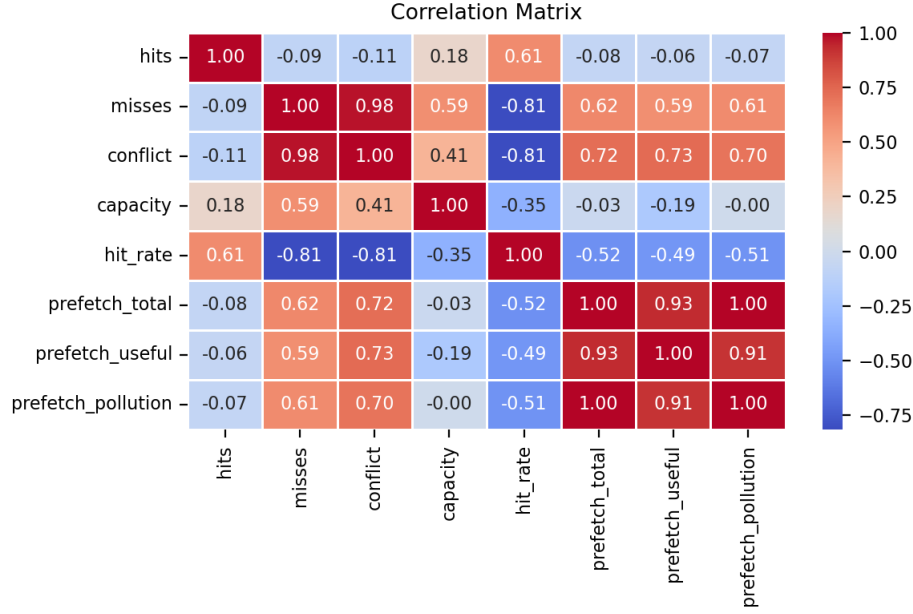


Figure 2: Correlation matrix of cache performance metrics.

Limitations This study is based on simulation and does not model full microarchitectural effects such as pipeline execution, out-of-order scheduling, or complete multi-level cache latency interactions. Additionally, the workloads used may not fully represent real-world application behaviour.

Future work

- Full multi-level cache hierarchy simulation (L1/L2/L3)
- Adaptive prefetching based on historical accuracy
- Replacement policy comparison (LRU vs FIFO vs Random)
- Evaluation using real memory trace datasets

Conclusion This project demonstrates that moderate associativity and conservative prefetching are jointly sufficient to capture most available cache performance. Four-way associativity eliminates the majority of conflict misses while avoiding the complexity of very high associativity, while prefetch distances of 1–2 consistently improve hit rate before pollution begins to dominate. The central finding is that these two mechanisms interact: higher associativity shifts the prefetch crossover point outward, meaning prefetch distance cannot be tuned independently of cache structure. Future work incorporating adaptive prefetching and multi-level hierarchies could exploit this interaction more precisely.

References

- [1] John L. Hennessy and David A. Patterson. *Computer Architecture: A Quantitative Approach*. 5th ed. Morgan Kaufmann, 2012.
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- [4] Alan Jay Smith. ‘Cache Memories’. In: *ACM Computing Surveys* 14.3 (1982).